March 16, 2026

# Open Source Hardware Hackathon 2026

# FARE-SHARE

https://oshw26.fossee.in/

# 1. Executive Summary

In many hostels and shared apartments, electricity bills are **split equally among all occupants**. However, electricity usage varies significantly between individuals depending on the appliances they use. Some residents consume minimal electricity, while others may use multiple high-power devices. Despite this difference, everyone pays the same amount.

*Fare Share* proposes a **low-cost IoT-based electricity monitoring system** that enables transparent electricity usage tracking in shared living spaces.

The system measures electricity consumption using sensors and sends the data to a **real-time dashboard**. Electricity costs are divided into:

- **Shared appliances** → split equally
- **Personal plug usage** → billed based on consumption

This approach enables **fair electricity billing without modifying existing building infrastructure**.

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# 2. Problem Statement

Electricity bills in shared living environments such as **hostels, PGs, and shared apartments** are usually divided equally among residents.

However, electricity consumption differs depending on the **number and type of appliances used by each individual**.

As a result, residents who consume less electricity often **pay the same amount as those who consume more**, leading to **unfair billing and lack of accountability in electricity usage**.

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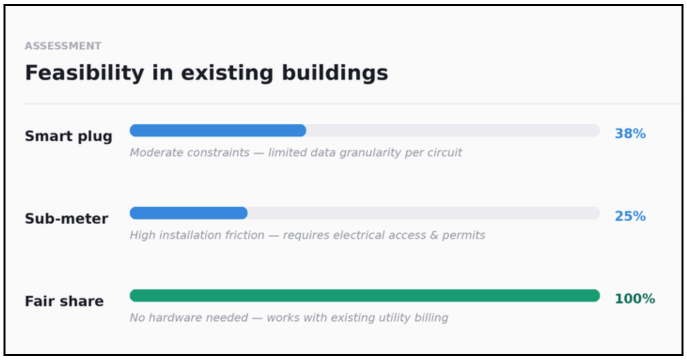
## 3. Market Gap

**Existing electricity monitoring solutions are not suitable for shared living spaces.**

| Existing Solution | Limitations |
| --- | --- |
| Smart Plug | Cannot monitor built-in appliances like fans, lights, AC |
| Sub-meter | Requires rewiring and infrastructure changes |

Because of these limitations, there is still a **lack of a simple, low-cost, and easily deployable solution** for monitoring electricity usage in shared living environments.

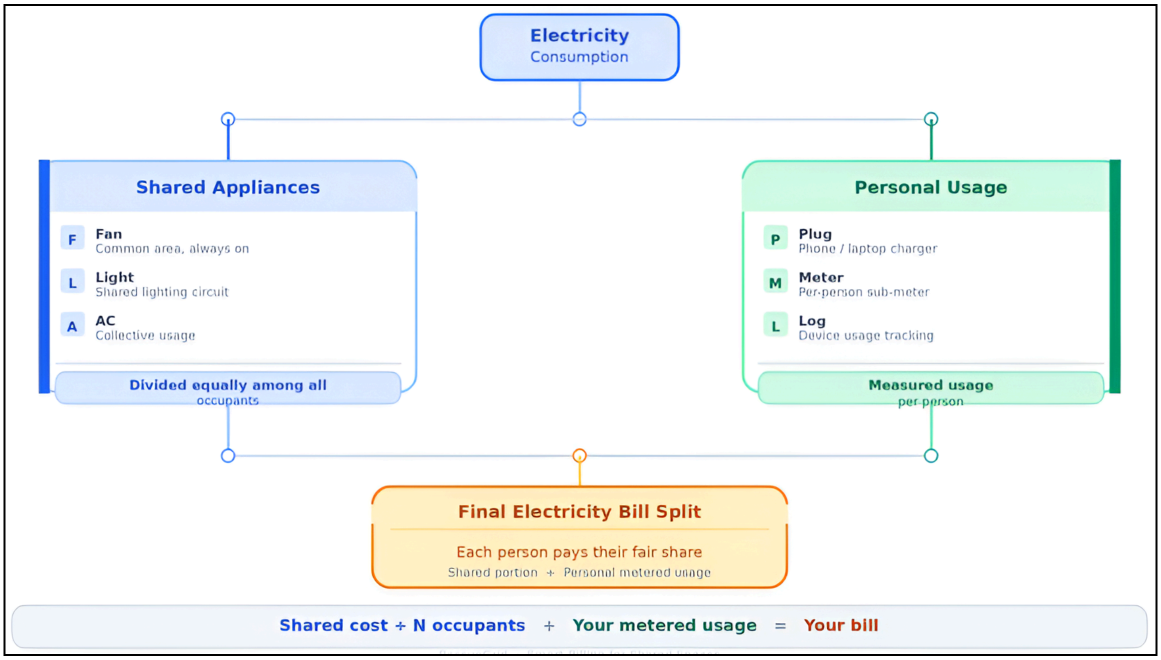



## 4. Objectives

The objectives of this project are:

- Develop a **low-cost electricity monitoring system**
- Enable **transparent electricity usage tracking**
- Create a **dashboard for real-time monitoring**
- Promote **fair electricity billing in shared spaces**

# 5. Fare Share: Concept Overview




By separating shared and personal consumption, *Fare Share* ensures that occupants **pay only for the electricity they actually use**, while shared appliances remain fairly distributed among everyone in the room.

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# 6. Literature Review

IoT-based systems are widely used to **monitor electricity consumption in buildings**. These systems typically use **current sensors, microcontrollers, and wireless communication** to collect energy usage data.

**Energy Consumption Before and After IoT Monitoring Implementation**

Measured in kWh · Before vs. After comparison across monitored rooms

| ROOM | BEFORE (KWH) | AFTER (KWH) | SAVED (KWH) | REDUCTION |
|---|---|---|---|---|
| Room A | 85.4 | 72.3 | -13.1 | ↓ 15.3% |
| Room B | 92.1 | 77.6 | -14.5 | ↓ 15.7% |
| Room C | 78.5 | 67.2 | -11.3 | ↓ 14.4% |
| Room D | 105.3 | 89.1 | -16.2 | ↓ 15.4% |

| TOTAL BEFORE | TOTAL AFTER | TOTAL SAVED | AVG REDUCTION |
|---|---|---|---|
| 361.3 kWh | 306.2 kWh | 55.1 kWh | 15.2% |

*Comparison of electricity consumption before and after implementing an IoT-based monitoring system across monitored rooms.*

Previous studies show that implementing IoT monitoring systems can **reduce electricity consumption by approximately 14–15%** through real-time monitoring and better energy awareness.

However, most existing systems focus on **building-level energy management**, while very few address the problem of **fair electricity cost distribution in shared living environments**.

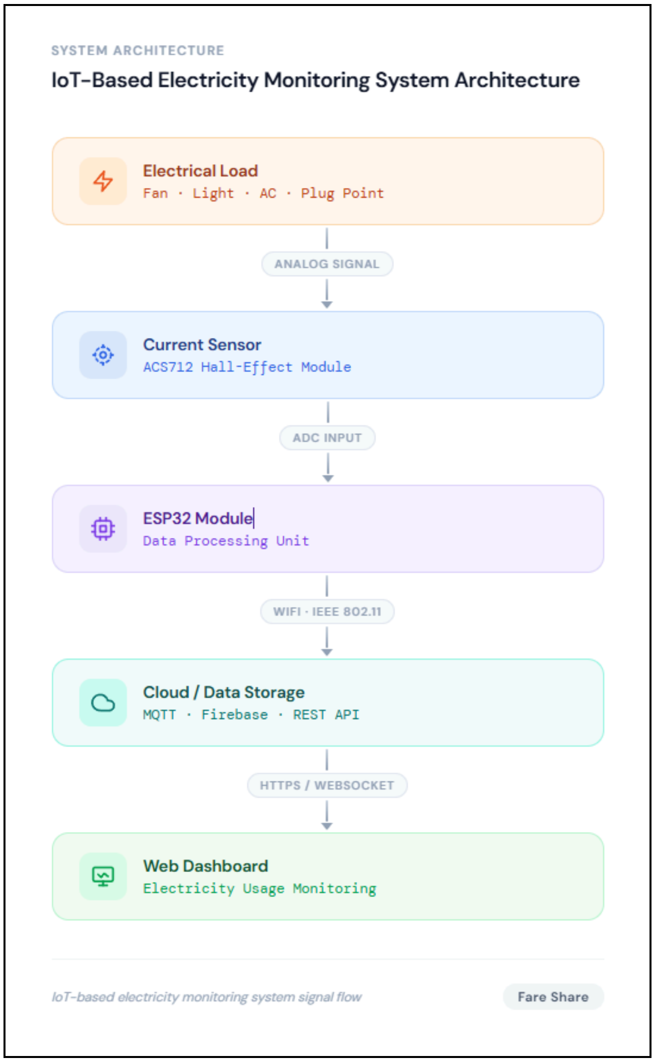
## 7. Proposed Solution

*Fare Share* is a **low-cost IoT electricity monitoring system** designed for shared living spaces.

The system works as follows:

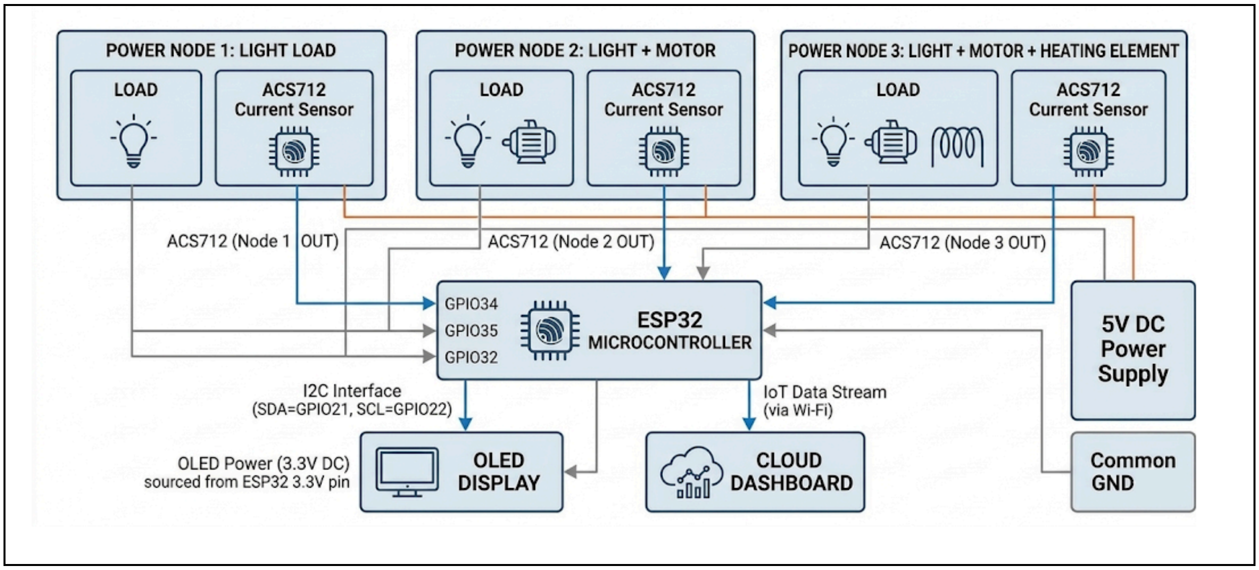
1. Electrical loads such as **fans, lights, AC, and plug points** draw power from the supply.
2. An **ACS712 current sensor** measures the current flowing through the circuit.
3. An **ESP microcontroller (ESP32/ESP8266)** reads the sensor data and processes it.
4. The processed data is sent via **Wi-Fi** to a **cloud dashboard**.
5. The dashboard displays **real-time electricity usage** for monitoring and billing.

An optional **OLED display** can show basic electrical data locally on the device.




This system enables **transparent monitoring and fair electricity cost distribution** without modifying existing electrical infrastructure.

# 8. Working Methodology




System operation steps:

1. Electrical appliances draw power from the supply.
2. Current and voltage sensors measure electrical parameters.
3. The microcontroller collects and processes the data.
4. Power consumption is calculated using:

$$P = V \times I$$

5. Data is transmitted to the IoT dashboard via Wi-Fi.
6. The dashboard displays **real-time electricity usage**.

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# 9. Results & Discussion

## Outcomes

- Low-cost IoT monitoring system
- Real-time electricity tracking
- Supports fair electricity cost distribution

## Limitations

- Sensor accuracy limitations
- Requires stable Wi-Fi

## Future Improvements

- Higher accuracy metering ICs
- Improved electrical isolation
- Alternative communication methods